Lander Zulueta

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Summary

Physics graduate and full-time experimental researcher in quantum sensing at Qubiz.team, building and operating a fiber-based quantum magnetometer (CW-ODMR, pulsed Rabi, T_1). Hands-on experience in microwave/RF engineering (waveguide design in HFSS and COMSOL, bench-top NMR), cryogenics and low-temperature STM at CIC nanoGUNE. Complemented by a thesis in quantum metrology.

Education

B.Sc. in Physics 2022 – 2026
University of the Basque Country (EHU) *Leioa, Spain*

- Grade: 9.06/10 in the Physics degree.
- Coursework: Quantum Mechanics, Thermodynamics, Statistical Mechanics, Electromagnetism, Solid State Physics, Advanced Laboratory Techniques.
- **Thesis** (10/10, Honors): *Critical Metrology* (see Research Experience).

Research Experience

Full-Time Researcher, Quantum Sensing Sep. 2026 – present
Qubiz.team *Bilbao, Spain*

- Built and operate a fiber-based optical quantum sensing setup for magnetic field sensing.
- Measured CW-ODMR (∼1.5% contrast), pulsed Rabi oscillations and T_1 relaxation.
- Designed and simulated microwave waveguides for hyperpolarization setups with Ansys HFSS and COMSOL RF.

Research Intern, Nanoimaging Group Jun. – Aug. 2026
CIC nanoGUNE (Supervisor: Prof. Nacho Pascual) *San Sebastián, Spain*

- Operated a low-temperature STM (LT-STM), including LHe/LN_2 transfers and sub-Kelvin cryostats.
- Characterized the transmission function of a cryogenic microwave antenna integrated into the STM junction, laying the RF calibration groundwork for ESR-STM measurements.
- Prepared STM tips and Au(111) substrates (UHV annealing, sputtering, MBE) and synthesized graphene nanoribbons by on-surface synthesis, verified by STM topography.

Researcher, Quantum Sensing Jun. – Dec. 2025
Qubiz.team *Bilbao, Spain*

- Contributed to early-stage projects in biomedicine, automotive and electrical industries, studying quantum sensing devices.
- Built RF components of a bench-top NMR device and tested classical machine-learning methods to enhance its signal.

Undergraduate Researcher (Bachelor's Thesis) Sep. 2025 – Jul. 2026
EHU (Supervisors: Iñigo Egusquiza, Jagoba Apellaniz; Ikasiker grant) *Leioa, Spain*

- *Critical Metrology.* Question: can quantum metrology gain from operating near a quantum phase transition? Method: used the superradiant transition of the quantum Rabi model in the classical-oscillator limit. Result: showed that the precision of parameter estimation diverges at the critical point in the static limit.
- Analytic work with Schrieffer-Wolff and bosonic Bogoliubov transformations; numerics with dense diagonalization and numerical differentiation.
- Code: github.com/Landerzl/Critical-Metrology

Selected Projects

- **BB84 QKD experiment.** Built an experiment at EHU to exchange messages with the BB84 protocol, including optical alignment and operation of the hardware.

Honors and Awards

- **Extraordinary Degree Award in Physics**, EHU (2026): best academic record of the 2025–26 cohort.
- **Ikasiker grant**, Basque Government (2026): competitive grant for the development of the Bachelor's Thesis.
- **Honors in 15 subjects** (2022–2026): Bachelor's Thesis (Critical Metrology), Solid State Physics I & II, Quantum Physics, Nuclear and Particle Physics, Experimental Techniques II, III & IV, Sensors and Actuators, Electromagnetism II, Modern Physics, Vector and Complex Analysis, Algebra I, Chemistry II, Fundamentals of Programming.
- **Academic Acceleration** (2021): skipped one year of secondary school for outstanding academic performance.

Courses and Certifications

- Quantum Machine Learning and Quantum Optimization (36h), University of Deusto, Nov. 2026 (upcoming)
- Laser Safety Officer (16h), ProCareLight, Sep. 2026
- Quantum Chemistry with VQE (5h), IBM, Sep. 2026
- Introduction to Ansys HFSS (2h), Ansys, Sep. 2026
- Quantum Communication and Quantum Networks (4h), TU Braunschweig, Feb. 2026
- General Formulation of Quantum Information (10h), IBM, Nov. 2025
- Fundamentals of Quantum Algorithms (10h), IBM, Aug. 2025
- Quantum Sensing and Imaging (30h), Qureca, Jul. 2025

Seminars and Workshops

- Hands on Quantum Information (3 ECTS), EHU Quantum Center, Mar. 2026 – present
- 8th Basque Quantum Science and Technology Workshop / 4th BasQ/IKUR Quantum Science and Technology Workshop, Oct. 2025
- Quantum Computing: from Fundamentals to Applications (1 ECTS), BasQ, DIPC, Sep. 2025
- Quantum Technologies: Present and Future, Nquire, Jun. 2025

Technical Skills

Experimental: Fiber-based quantum sensing (CW-ODMR, Rabi, T_1), microwave/RF delivery, cryogenics, UHV and low-temperature STM
Programming: Python (QuTiP, scqubits, PySCF, NumPy, SciPy), MATLAB, Wolfram Mathematica, Fortran 90
Simulation and RF: COMSOL Multiphysics, Ansys HFSS, Qorvo MatchCalc, Smith charts, LAMMPS, Fusion 360, PSpice
Tools: LaTeX, Git/GitHub, LabVIEW, Linux (WSL)
Languages: Basque (native), Spanish (native), English (C1, Cambridge, Sep. 2025), French (basic)

References

Prof. Nacho Pascual, CIC nanoGUNE (internship supervisor)
Prof. Iñigo Egusquiza, University of the Basque Country (EHU) (thesis supervisor)